

Chapter 15

Chi-Square, Weibull, Lognormal, Beta

Lecture 15 | Devore 9e: 4.4–4.5

Last time: change of variable, exponential, gamma. Today we finish the gamma and collect the remaining named continuous families. Every one of them is built out of pieces we already have.

15.1 Gamma, continued

Devore writes the gamma density in the equivalent form

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta \Gamma(\alpha)} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-x/\beta}, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad \mathbb{E}[X] = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2.$$

Proposition 15.1 (Scaling). *If $X \sim \text{Gamma}(\alpha, \beta)$ then $\frac{X}{\beta} \sim \text{Gamma}(\alpha, 1)$.*

This is the change of variable of Lecture 14 with $g(x) = x/\beta$: the inverse is $g^{-1}(u) = \beta u$ with $\left|\frac{d}{du}g^{-1}(u)\right| = \beta$, so

$$f_{X/\beta}(u) = \beta f_X(\beta u) = \beta \cdot \frac{(\beta u)^{\alpha-1} e^{-u}}{\beta^\alpha \Gamma(\alpha)} = \frac{u^{\alpha-1} e^{-u}}{\Gamma(\alpha)}.$$

So, just as with the normal, one table suffices.

Definition 15.2 (Incomplete gamma function). For the *standard* gamma ($\beta = 1$),

$$\text{IGF}(x; \alpha) := \mathbb{P}\{X \leq x\} = \frac{1}{\Gamma(\alpha)} \int_0^x y^{\alpha-1} e^{-y} dy.$$

Then for general β , by the scaling property,

$$\mathbb{P}\{X \leq x\} = \mathbb{P}\left\{\frac{X}{\beta} \leq \frac{x}{\beta}\right\} = \text{IGF}\left(\frac{x}{\beta}; \alpha\right),$$

which is looked up in a table exactly as Φ was.

Theorem 15.3 (Additivity). *If $X_1 \sim \text{Gamma}(\alpha_1, \beta)$ and $X_2 \sim \text{Gamma}(\alpha_2, \beta)$ are indepen-*

dent (same $\beta!$), then

$$X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta).$$

This is just the waiting-time reading: waiting for α_1 arrivals and then for α_2 more is waiting for $\alpha_1 + \alpha_2$ arrivals. It also explains what a *fractional* α means: any $\text{Gamma}(\alpha, \beta)$ splits as a sum of a $\text{Gamma}(\lfloor \alpha \rfloor, \beta)$ piece and a $\text{Gamma}(\alpha - \lfloor \alpha \rfloor, \beta)$ piece.

Remark 15.4 (Shape). $0 < \alpha < 1$ gives a density with a spike at 0 — the “early failure” regime, appropriate for items that fail immediately or not at all. $\alpha > 1$ gives an interior mode.

15.2 The chi-square distribution

Definition 15.5 (Chi-square). Let $Z_1, \dots, Z_k$ be i.i.d. $\text{Normal}(0, 1)$ and set

$$Y = Z_1^2 + Z_2^2 + \dots + Z_k^2.$$

Then $Y \sim \chi_k^2$, chi-square with k **degrees of freedom**, with pdf

$$f(y; k) = \begin{cases} \frac{1}{2^{k/2} \Gamma(k/2)} y^{k/2-1} e^{-y/2}, & y > 0, \\ 0, & y \leq 0. \end{cases}$$

Comparing with the gamma density:

$$\boxed{\chi_k^2 = \text{Gamma}\left(\frac{k}{2}, 2\right)} \implies \mathbb{E}[Y] = \frac{k}{2} \cdot 2 = k, \quad \text{Var}(Y) = \frac{k}{2} \cdot 2^2 = 2k.$$

(For $k = 1$ this matches the transformation $Y = Z^2$ we did in Lecture 14.)

Remark 15.6 (Intuition: energy). A gas molecule has velocity $\vec{V} = (V_x, V_y, V_z)$ with each component approximately $\text{Normal}(0, 1)$ after scaling. Its kinetic energy is proportional to

$$V_x^2 + V_y^2 + V_z^2 \sim \chi_3^2.$$

Chi-square is what you get when you add up squared fluctuations — which is exactly what a sample variance does, and why this distribution runs the statistics half of the course.

15.3 The Weibull distribution

Definition 15.7 (Weibull). $X \sim \text{Weibull}(\alpha, \beta)$ with shape $\alpha > 0$ and scale $\beta > 0$ if

$$f(x; \alpha, \beta) = \begin{cases} \frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-(x/\beta)^\alpha}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Its cdf is unusually simple:

$$F(x) = 1 - e^{-(x/\beta)^\alpha}, \quad x \geq 0.$$

Remark 15.8 (Weibull versus gamma — look at the exponent).

$$\begin{aligned}\text{Weibull: } & \frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^{\alpha}} \\ \text{Gamma: } & \frac{1}{\beta\Gamma(\alpha)} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)}\end{aligned}$$

The polynomial factor is the same and the constants out front differ only in what it takes to normalize; the families differ in whether α appears in the exponent of the exponential. That one change is what makes the Weibull cdf elementary while the gamma cdf requires a table.

Theorem 15.9 (Where it comes from). *If $T \sim \text{Exponential}(1)$ and $X := \beta T^{1/\alpha}$, then $X \sim \text{Weibull}(\alpha, \beta)$.*

Proof. $\mathbb{P}\{X \leq x\} = \mathbb{P}\{\beta T^{1/\alpha} \leq x\} = \mathbb{P}\{T \leq (x/\beta)^\alpha\} = 1 - e^{-(x/\beta)^\alpha}$. $\square$

So the Weibull is a *stretched* exponential waiting time. Setting $\alpha = 1$ recovers $\text{Exponential}(1/\beta)$ exactly.

Theorem 15.10.

$$\mathbb{E}[X] = \beta \Gamma\left(1 + \frac{1}{\alpha}\right), \quad \text{Var}(X) = \beta^2 \left[\Gamma\left(1 + \frac{2}{\alpha}\right) - \Gamma\left(1 + \frac{1}{\alpha}\right)^2 \right].$$

Remark 15.11 (Why engineers use it). The exponential is memoryless, so it cannot describe wear-out. The Weibull can, through α :

$\alpha < 1$ failure rate *decreasing* — infant mortality
 $\alpha = 1$ constant failure rate — exponential, no aging
 $\alpha > 1$ failure rate *increasing* — wear-out

This is the standard lifetime model in reliability engineering.

15.4 The lognormal distribution

Definition 15.12 (Lognormal). $X \sim \text{Lognormal}(\mu, \sigma^2)$ if $\ln X \sim \text{Normal}(\mu, \sigma^2)$.

Derive the density by the usual cdf route: for $x > 0$,

$$\mathbb{P}\{X \leq x\} = \mathbb{P}\{\ln X \leq \ln x\} = \Phi\left(\frac{\ln x - \mu}{\sigma}\right),$$

and differentiating (chain rule, $\frac{d}{dx} \ln x = \frac{1}{x}$),

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} \cdot \frac{1}{x}, \quad x > 0.$$

Theorem 15.13.

$$\mathbb{E}[X] = e^{\mu + \sigma^2/2}, \quad \text{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}.$$

Remark 15.14 (Warning). μ and σ^2 are the mean and variance of $\ln X$, *not* of X . Note $\mathbb{E}[X] = e^{\mu + \sigma^2/2} > e^\mu$, which is the median. The lognormal is right-skewed.

Example 15.15 (Multiplicative growth). A stock grows by a random percentage each day:

$$X = X_0(1 + R_1)(1 + R_2) \cdots (1 + R_n).$$

Take logs:

$$\ln X = \ln X_0 + \sum_{i=1}^n \ln(1 + R_i).$$

The right-hand side is a *sum* of many small independent terms, so by the central limit theorem it is approximately normal — and therefore X itself is approximately lognormal.

Remark 15.16. So: normal is what you get from adding many small effects; lognormal is what you get from *multiplying* them. Prices, incomes, particle sizes, and file sizes are lognormal for this reason.

Example 15.17 (Mean versus median). One dollar invested today is worth $W = e^Y$ dollars in ten years, where $Y \sim \text{Normal}(0, 0.9)$. (Ten independent years, each with a $\text{Normal}(0, 0.09)$ log-return, add up to this; Lecture 20 shows why.) What is the typical outcome, and what is the average one?

The median of W is $e^0 = 1$: since $\mathbb{P}\{W \leq 1\} = \mathbb{P}\{Y \leq 0\} = \frac{1}{2}$, it is a coin flip whether the investor ends up with more than they put in. Yet

$$\mathbb{E}[W] = e^{0+0.9/2} = e^{0.45} \approx 1.57,$$

an average gain of 57%. The average is carried by a few very large outcomes in the long right tail; the typical outcome is no gain at all.

Remark 15.18 (Volatility drag). Here is the same effect in its simplest form. A price that rises 50% and then falls 50% ends at $1.5 \times 0.5 = 0.75$ of where it started, although the average of the two returns is zero. Under multiplicative growth, randomness pulls the median below the mean, and more volatility pulls it further: the ratio is $e^{\sigma^2/2}$.

15.5 The beta distribution

Everything so far has had unbounded support. For a random *proportion* we need a distribution living on $[0, 1]$.

Definition 15.19 (Standard beta). $X \sim \text{Beta}(\alpha, \beta)$ if

$$f(x; \alpha, \beta) = \begin{cases} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

To place it on a general interval $[A, B]$: say Y is beta on $[A, B]$ if $\frac{Y-A}{B-A} \sim \text{Beta}(\alpha, \beta)$.

Theorem 15.20.

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

Theorem 15.21 (Where it comes from). *Let $Y_1 \sim \text{Gamma}(\alpha, 1)$ and $Y_2 \sim \text{Gamma}(\beta, 1)$ be independent. Then*

$$X := \frac{Y_1}{Y_1 + Y_2} \sim \text{Beta}(\alpha, \beta).$$

Remark 15.22. That is the right way to read the parameters: α and β are the “sizes” of two competing gamma quantities, and X is the *share* taken by the first. For example, if spring rainfall is $\text{Gamma}(\alpha, 1)$ and the rest of the year is $\text{Gamma}(\beta, 1)$, then the fraction of the annual total falling in spring is $\text{Beta}(\alpha, \beta)$. Note $\text{Beta}(1, 1) = \text{Uniform}[0, 1]$.

15.5.1 Ordered uniforms

The beta family also answers a question left open in Lecture 12. Drop n points uniformly on $[0, 1]$ by independent trials, and let $U_{(k)}$ be the k -th smallest. How is $U_{(k)}$ distributed?

Theorem 15.23 (Order statistics of uniforms). $U_{(k)} \sim \text{Beta}(k, n - k + 1)$. *In particular*

$$\mathbb{E}[U_{(k)}] = \frac{k}{n + 1}.$$

Proof. $U_{(k)} \leq x$ exactly when at least k of the n points fall in $[0, x]$, and the number that do is $\text{Binomial}(n, x)$. So

$$F(x) = \sum_{j=k}^n \binom{n}{j} x^j (1-x)^{n-j}, \quad 0 \leq x \leq 1.$$

Differentiate term by term. Using $j \binom{n}{j} = n \binom{n-1}{j-1}$ and $(n-j) \binom{n}{j} = n \binom{n-1}{j}$,

$$\frac{d}{dx} \left[\binom{n}{j} x^j (1-x)^{n-j} \right] = a_j - a_{j+1}, \quad a_j := n \binom{n-1}{j-1} x^{j-1} (1-x)^{n-j},$$

with $a_{n+1} := 0$. The sum telescopes to

$$f(x) = a_k = n \binom{n-1}{k-1} x^{k-1} (1-x)^{n-k} = \frac{n!}{(k-1)!(n-k)!} x^{k-1} (1-x)^{n-k},$$

which is the $\text{Beta}(k, n - k + 1)$ density, because $\Gamma(m) = (m-1)!$. The mean is $\frac{k}{k+(n-k+1)} = \frac{k}{n+1}$. $\square$

The density also has a one-line reading. For $U_{(k)}$ to land in a short interval $[x, x + dx]$, one of the n points must land there, $k-1$ of the other $n-1$ must fall below x , and the remaining $n-k$ above:

$$n \binom{n-1}{k-1} x^{k-1} (1-x)^{n-k} dx.$$

For $k = n$ and $k = 1$ this recovers the maximum and the minimum of Lecture 12, and the mean $\frac{k}{n+1}$ is the picture from there: k gaps, each of expected length $\frac{1}{n+1}$.

Example 15.24 (The median of five). Five numbers are drawn uniformly from $[0, 1]$. Their median is $U_{(3)} \sim \text{Beta}(3, 3)$, with density $30x^2(1-x)^2$, mean $\frac{1}{2}$, and variance $\frac{3 \cdot 3}{6^2 \cdot 7} = \frac{1}{28}$, so a standard deviation of about 0.19.

15.6 Summary: the continuous families

Distribution	Relation	What it models
Uniform $[A, B]$	—	equally likely anywhere in an interval
Normal (μ, σ^2)	—	sums of many small independent effects
Exponential (λ)	$= \text{Gamma}(1, 1/\lambda)$	waiting time for the 1st arrival; memoryless lifetime
Gamma (α, β)	—	waiting time for α arrivals
χ_k^2	$= \text{Gamma}(k/2, 2)$	sum of k squared standard normals; “energy”
Weibull (α, β)	$\beta T^{1/\alpha}, T \sim \text{Exponential}(1)$	stretched waiting time; lifetimes <i>with</i> aging
Lognormal (μ, σ^2)	$\ln X \sim \text{Normal}(\mu, \sigma^2)$	multiplicative growth
Beta (α, β)	$\frac{Y_1}{Y_1 + Y_2}, Y_i$ gamma	a random proportion in $[0, 1]$; the k -th smallest of n uniforms

Remark 15.25. Notice how few genuinely independent objects there are. Starting from a Poisson process you get exponential, gamma, chi-square, Weibull and beta; starting from the normal you get chi-square and lognormal. Learning the *relations* is far more economical than memorizing eight densities.